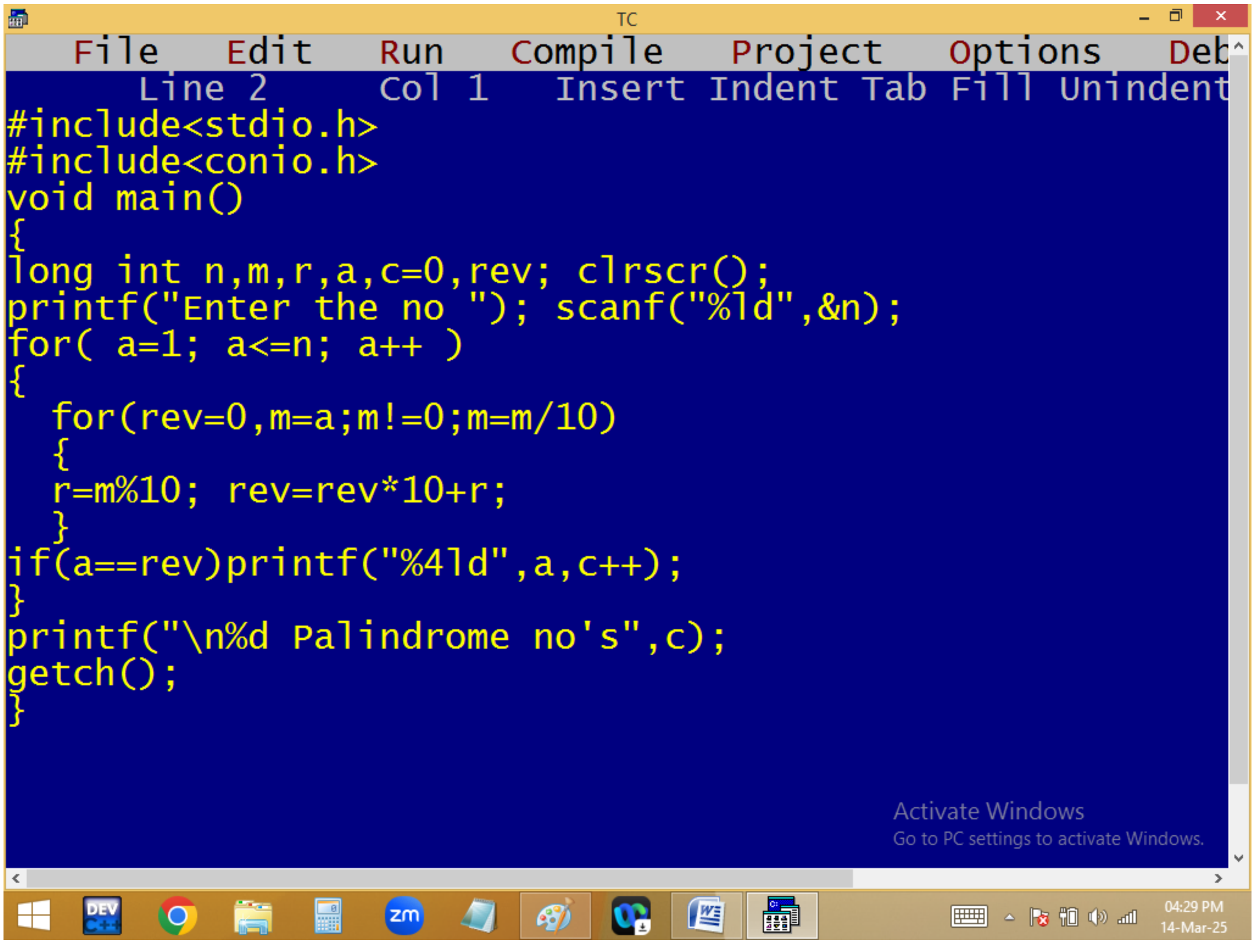
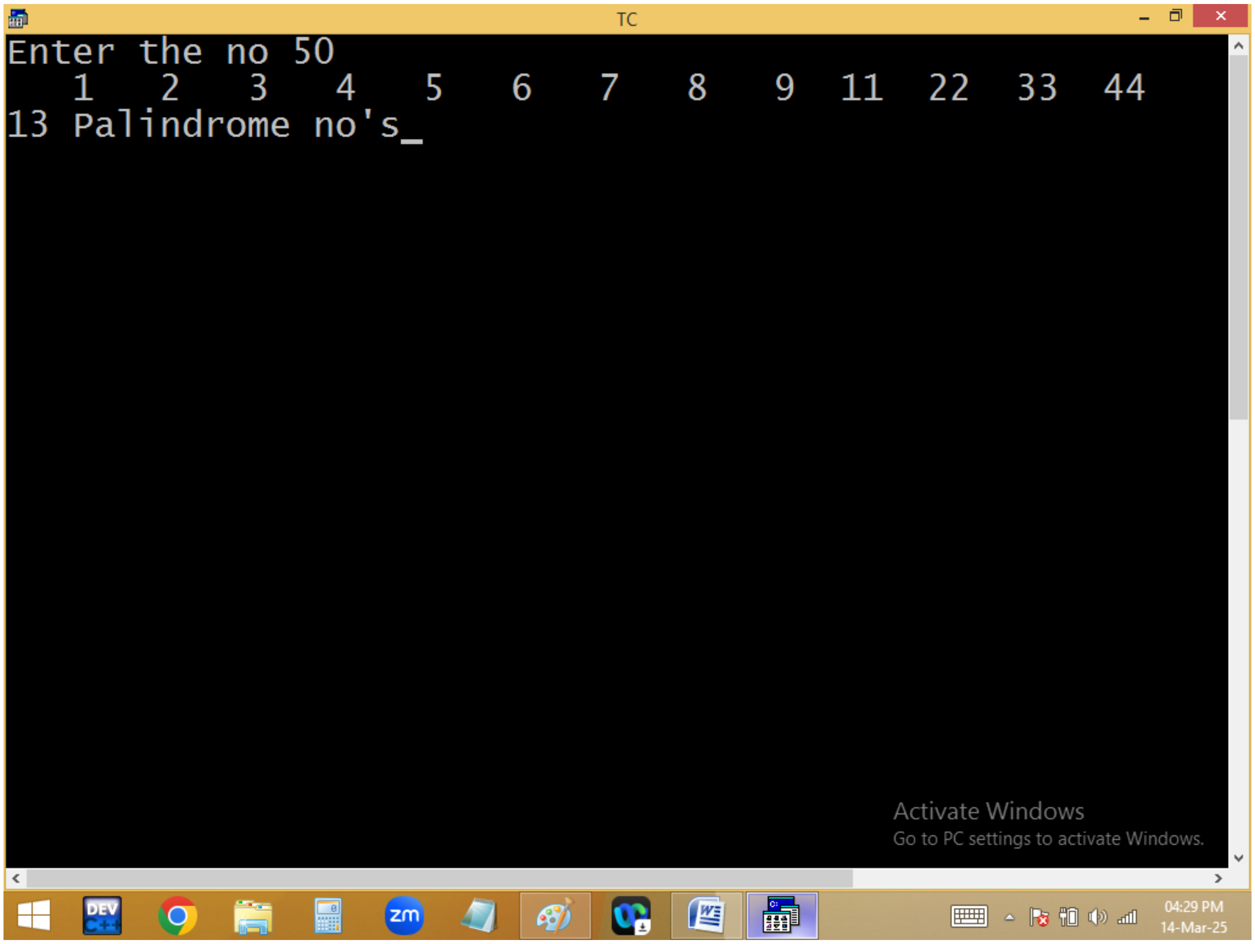
Printing 1..n palindrome no’s and count




```c
#include<stdio.h>
#include<conio.h>
void main()
{
long int n,m,r,a,c=0,rev; clrscr();
printf("Enter the no "); scanf("%ld",&n);
for( a=1; a<=n; a++ )
{
  for(rev=0,m=a;m!=0;m=m/10)
  {
  r=m%10; rev=rev*10+r;
  }
if(a==rev)printf("%4ld",a,c++);
}
printf("\n%d Palindrome no's",c);
getch();
}
```



a to n → m r rev

1

313 ~~313~~%10=3 0x10+3=3

~~31~~%10=1 3x10+1=31

~~3~~%10=3 31x10+3=313

0

```
for( a=1; a<=n; a++ )
{       rev=0;
 for( m=a; m!=0; m=m/10 )
 {
 r=m%10;
 rev=rev*10+r;
 }
 puts(a==rev)p(palin);

 }
```

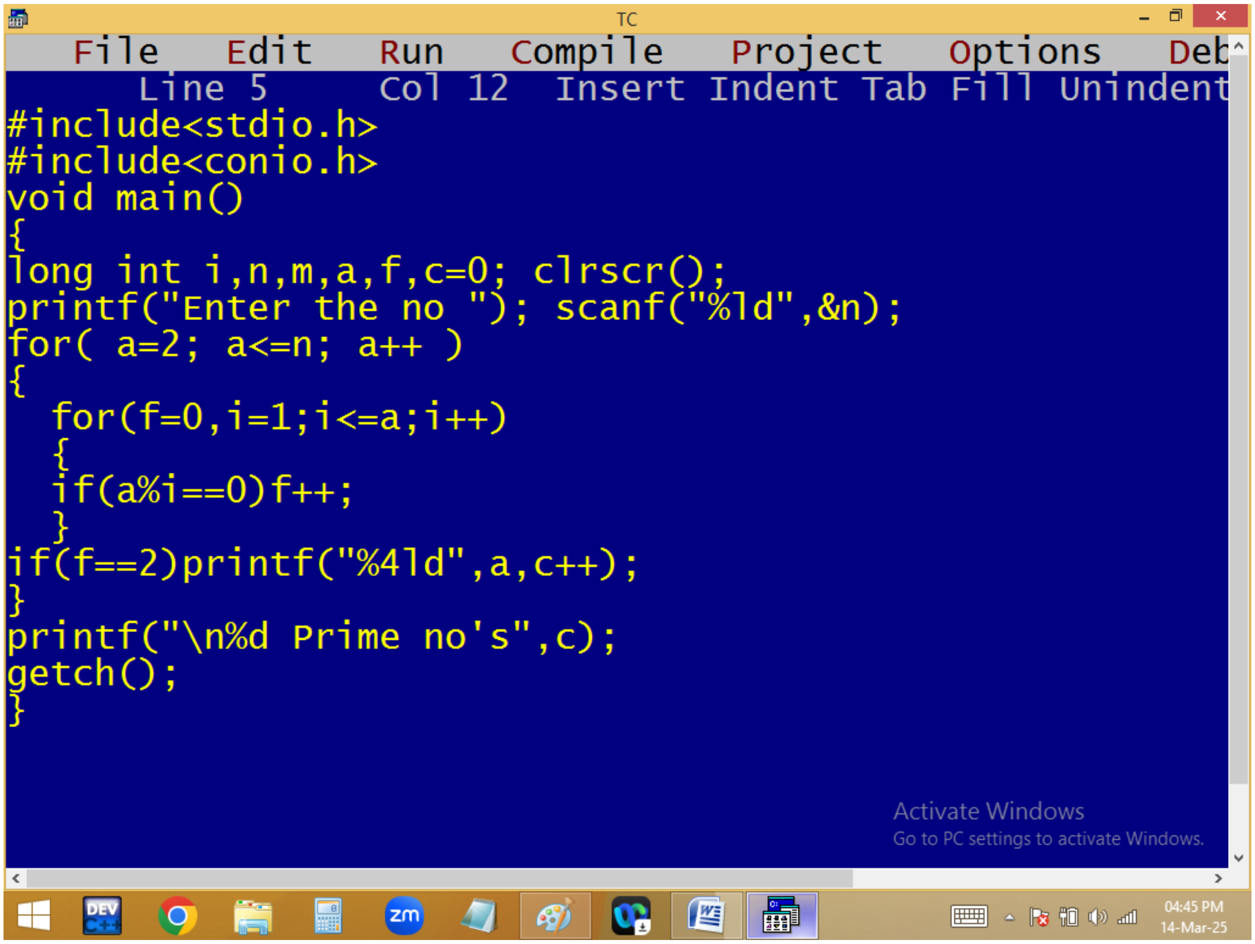
```
for( a=1; a<=n; a++ )
{
      rev=0;
 for( m=a; m!=0; m=m/10 )
 {
 r=m%10;
 rev=rev*10+r;
 }
 puts(a==rev)p(palin);

 }
```

n	a	m	r	rev	c
50	✓ 1	0	1%10=1	0x10+1=1	1
	✓ 2	0	2%10= 2	0x10+2=2	2
					9
	✗ 10		10%10 = 0	0x10+0=0	
		0	1%10 = 1	0x10+1=1	
	✓ 11		11%10=1	0x10+1=1	10
		0	1%10 =1	1x10+1=11	

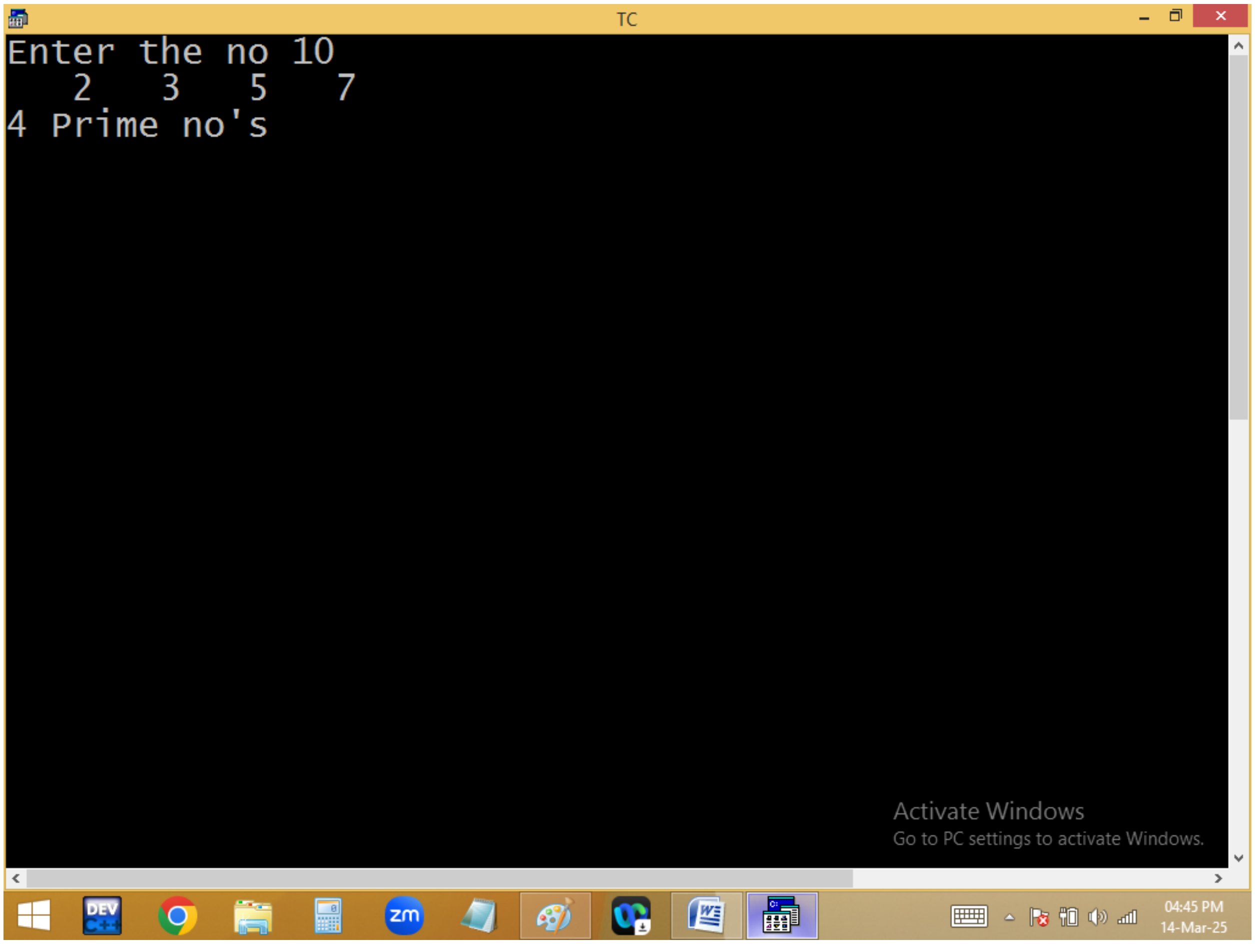
Printing 1..n primes and count

```
for( i=1; i<=3; i++ )
{
if(n%i==0)c++;
}
if(c==2) p(prime);
```

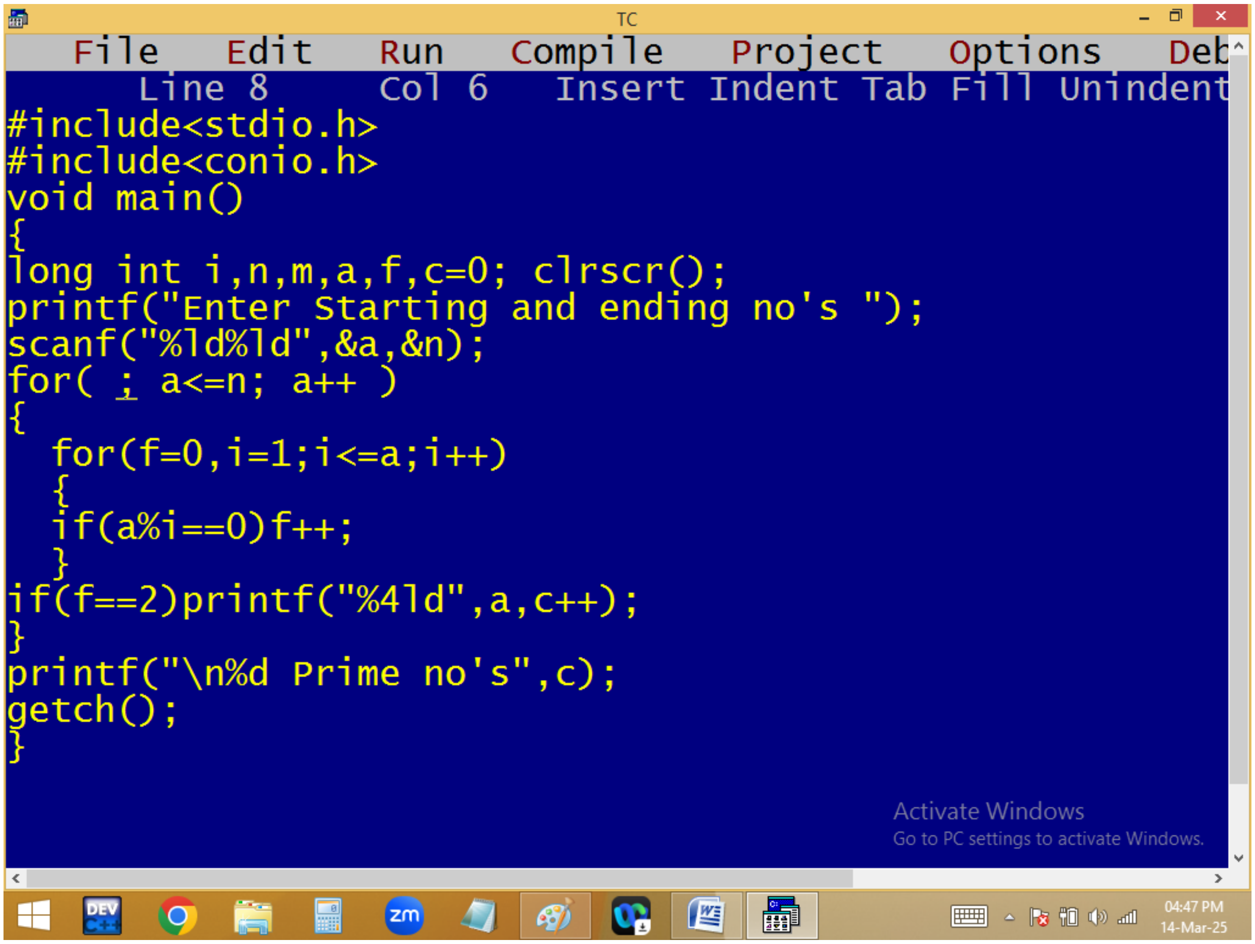
n	i		c
3 %	1 =	0	1
3 %	2 =	1	
3 %	3 =	0	2



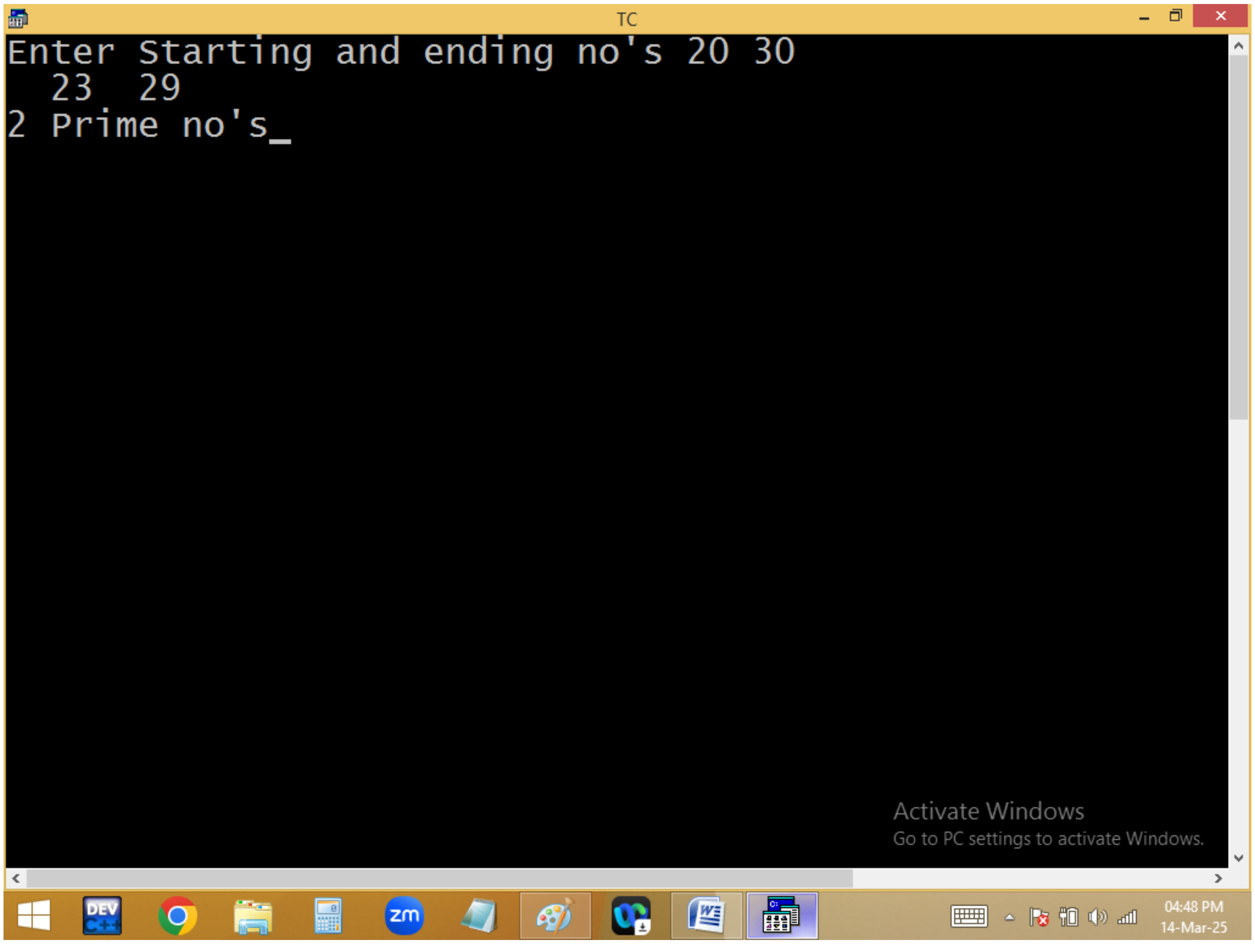

```c
#include<stdio.h>
#include<conio.h>
void main()
{
long int i,n,m,a,f,c=0; clrscr();
printf("Enter the no "); scanf("%ld",&n);
for( a=2; a<=n; a++ )
{
  for(f=0,i=1;i<=a;i++)
  {
  if(a%i==0)f++;
  }
if(f==2)printf("%4ld",a,c++);
}
printf("\n%d Prime no's",c);
getch();
}
```

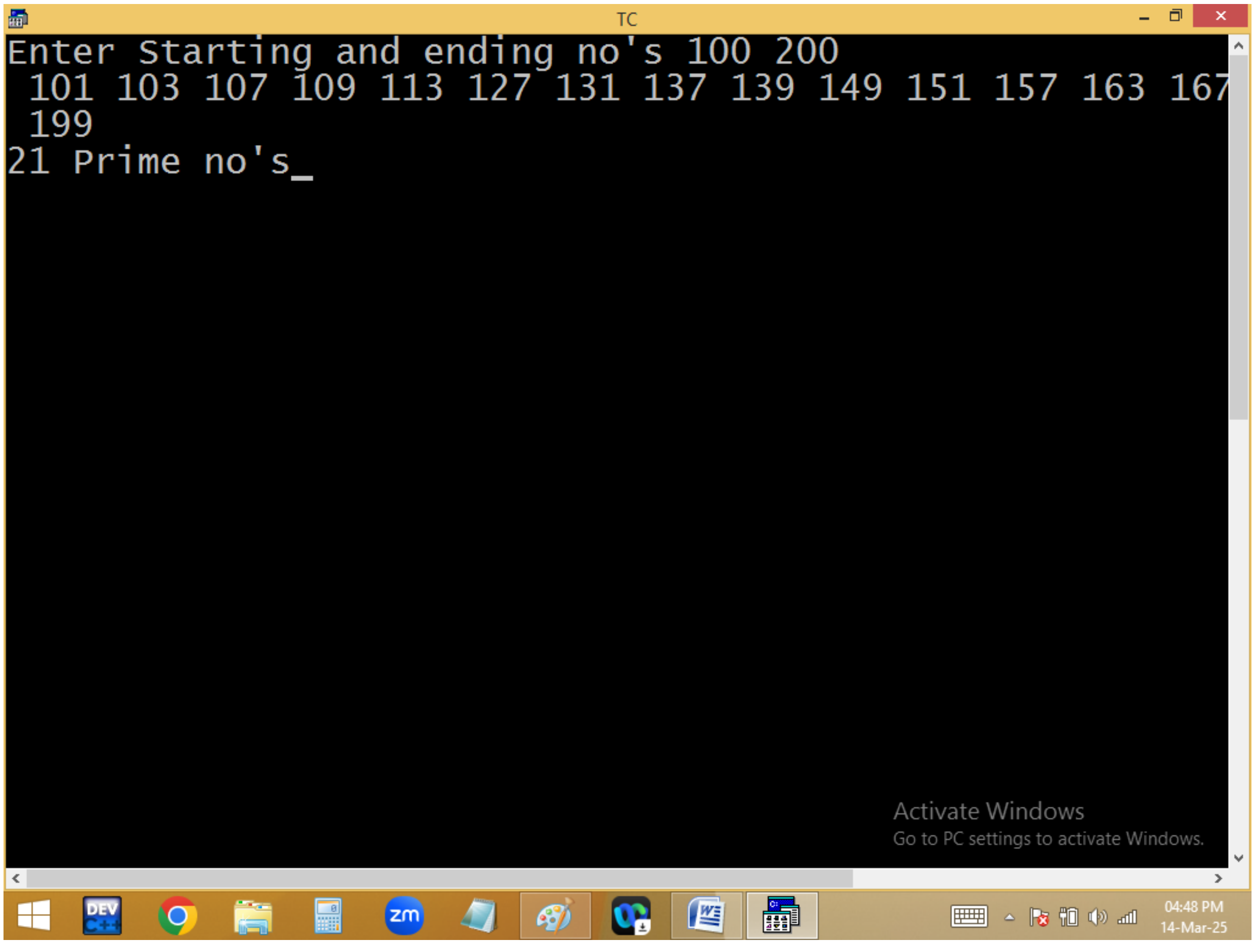


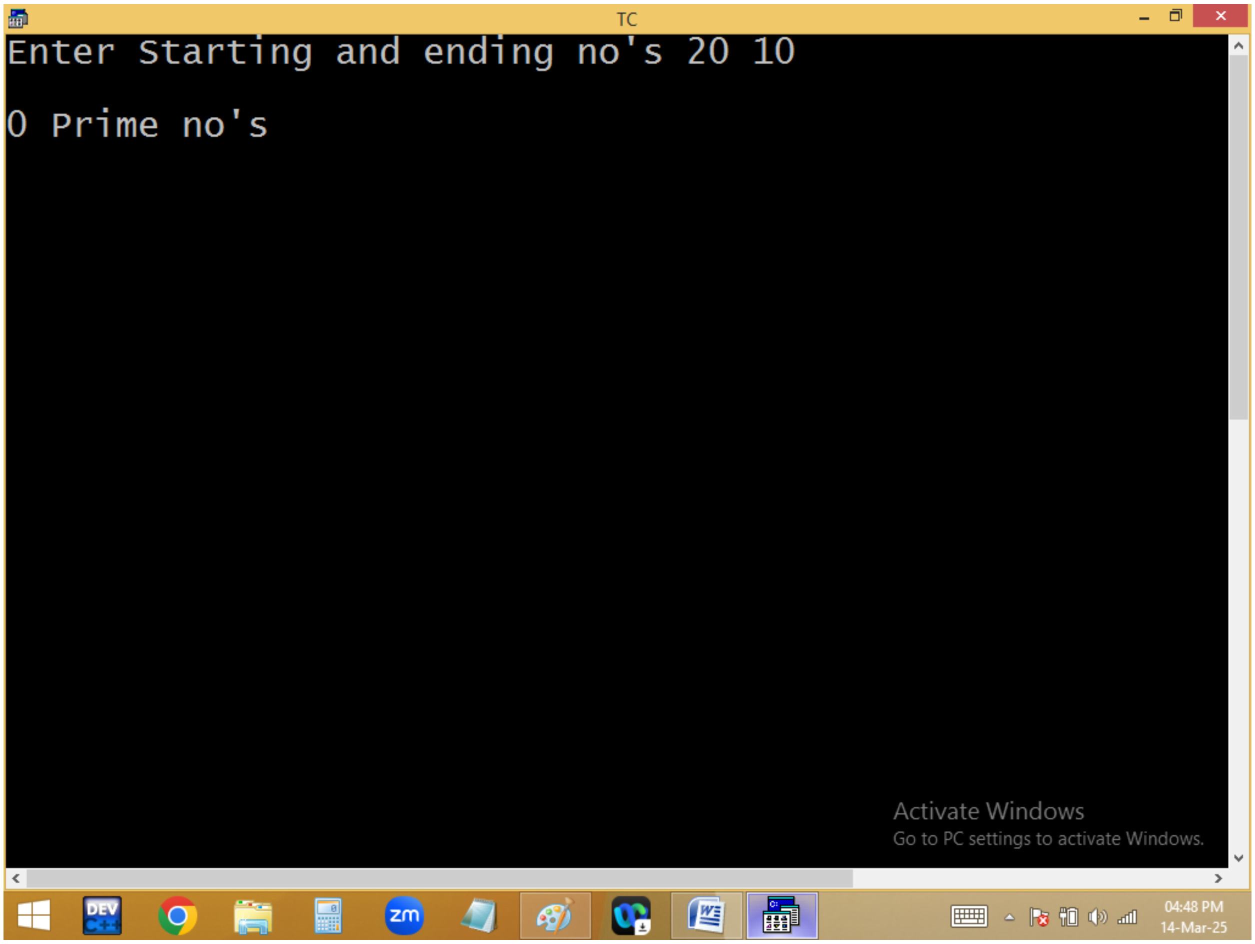
Printing n..n primes and count

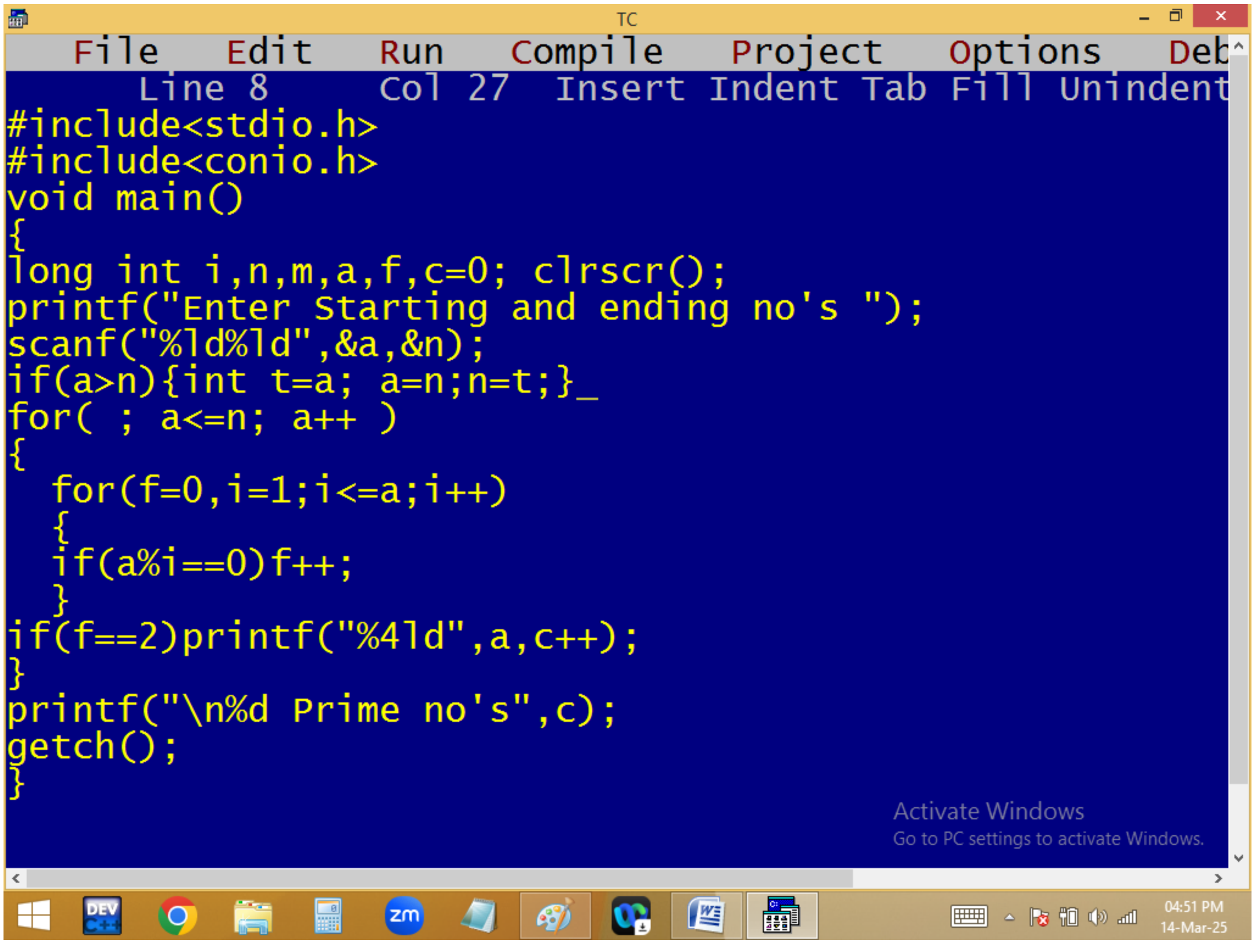



```c
#include<stdio.h>
#include<conio.h>
void main()
{
long int i,n,m,a,f,c=0; clrscr();
printf("Enter Starting and ending no's ");
scanf("%ld%ld",&a,&n);
for( ; a<=n; a++ )
{
  for(f=0,i=1;i<=a;i++)
  {
  if(a%i==0)f++;
  }
if(f==2)printf("%4ld",a,c++);
}
printf("\n%d Prime no's",c);
getch();
}
```

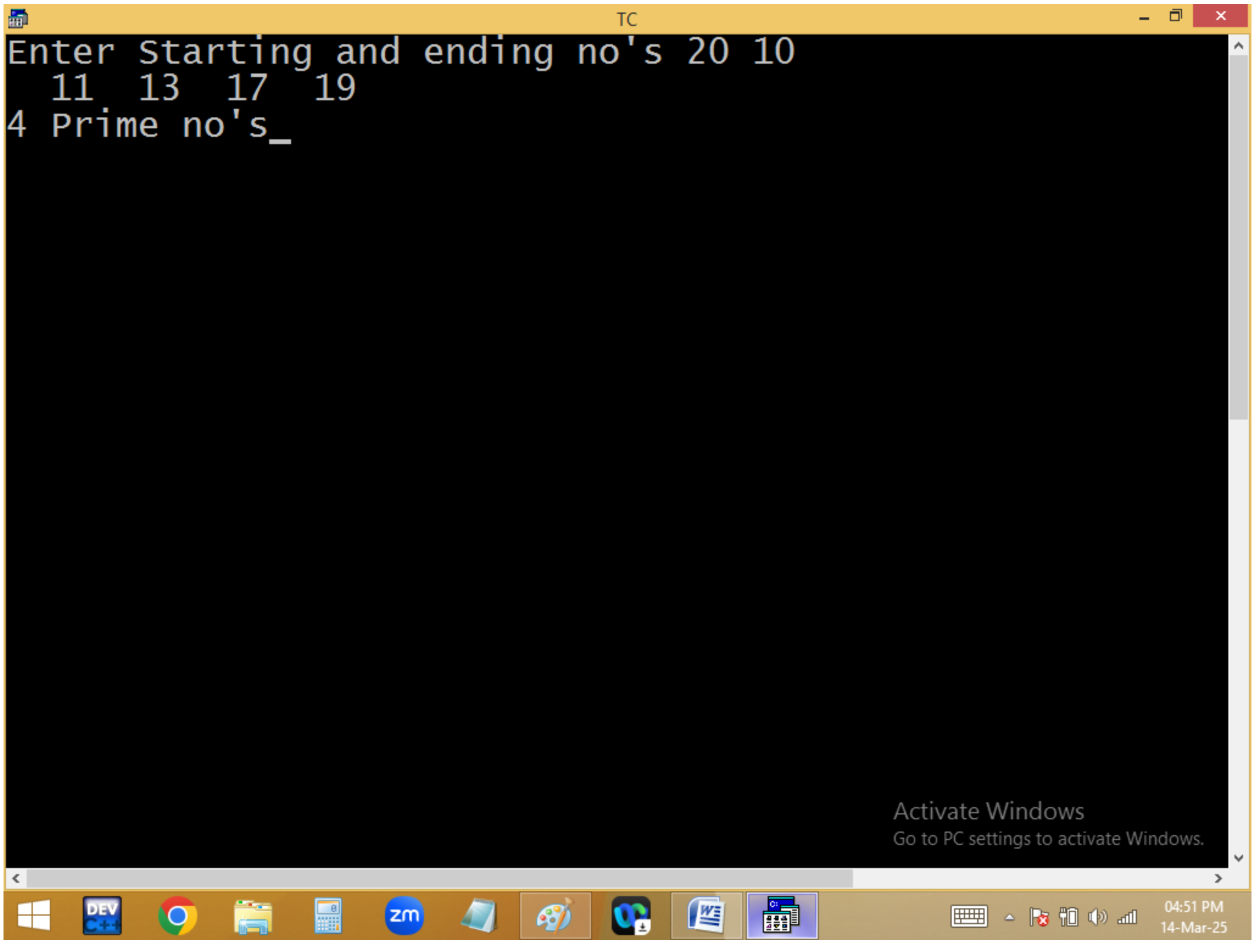

TC
Enter Starting and ending no's 20 30
23 29
2 Prime no's_
Activate Windows
Go to PC settings to activate Windows.
04:48 PM
14-Mar-25


TC
Enter Starting and ending no's 100 200
101 103 107 109 113 127 131 137 139 149 151 157 163 167
199
21 Prime no's_
Activate Windows
Go to PC settings to activate Windows.
04:48 PM
14-Mar-25


TC
Enter Starting and ending no's 20 10
0 Prime no's
Activate Windows
Go to PC settings to activate Windows.
04:48 PM
14-Mar-25



```
#include<stdio.h>
#include<conio.h>
void main()
{
long int i,n,m,a,f,c=0; clrscr();
printf("Enter Starting and ending no's ");
scanf("%ld%ld",&a,&n);
if(a>n){int t=a; a=n;n=t;}_
for( ; a<=n; a++ )
{
  for(f=0,i=1;i<=a;i++)
  {
  if(a%i==0)f++;
  }
if(f==2)printf("%4ld",a,c++);
}
printf("\n%d Prime no's",c);
getch();
}
```

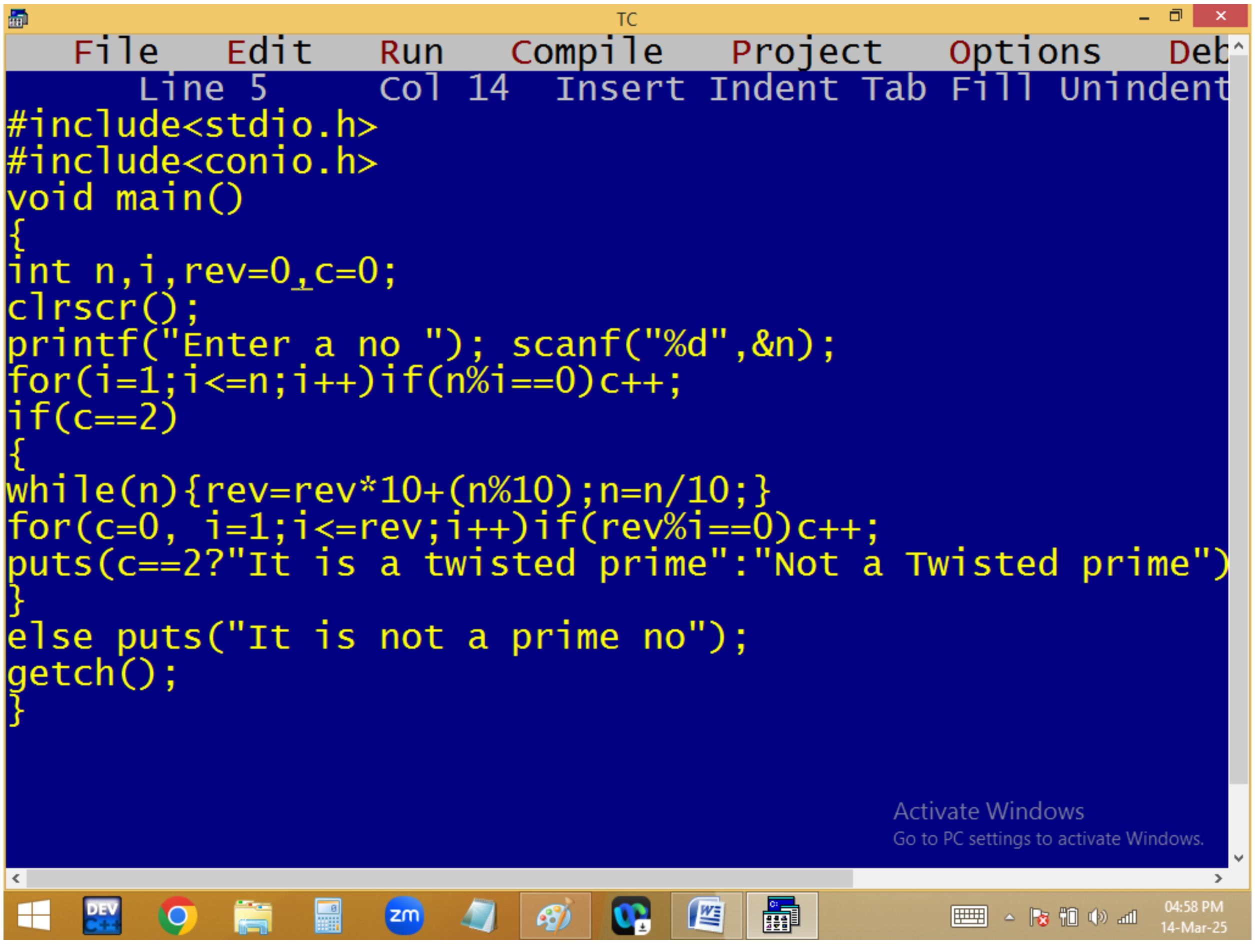


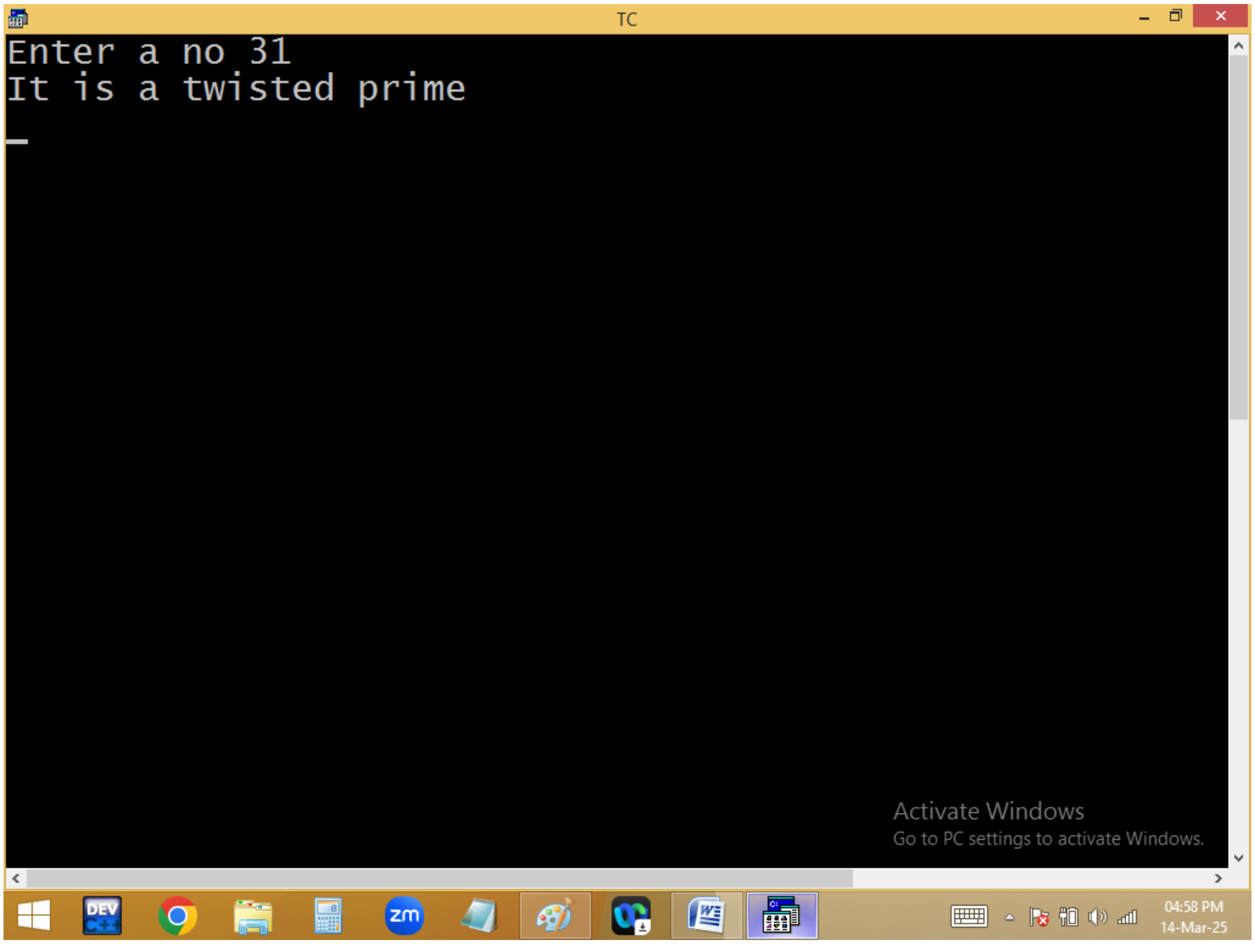

```
for( a=2;a<=n; a++)
{       c=0;
   for( i=1; i<=a; i++ )
   {
   if(a %i==0)c++;
   }
   if(c==2) p(prime);

}
p(c);
```

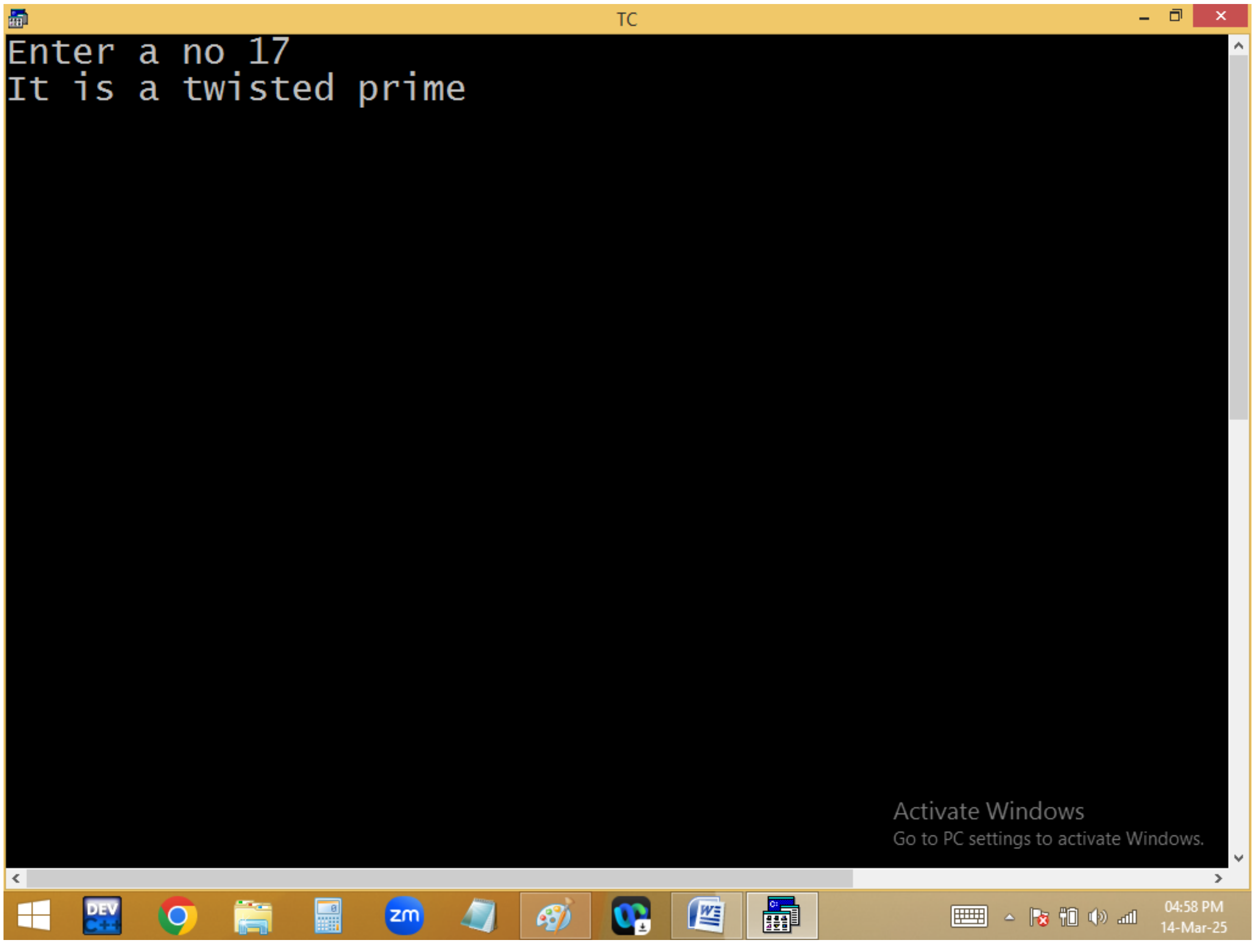
n	i	c
3 % 1 = 0		1
3 % 2 = 1		
3 % 3 = 0		2

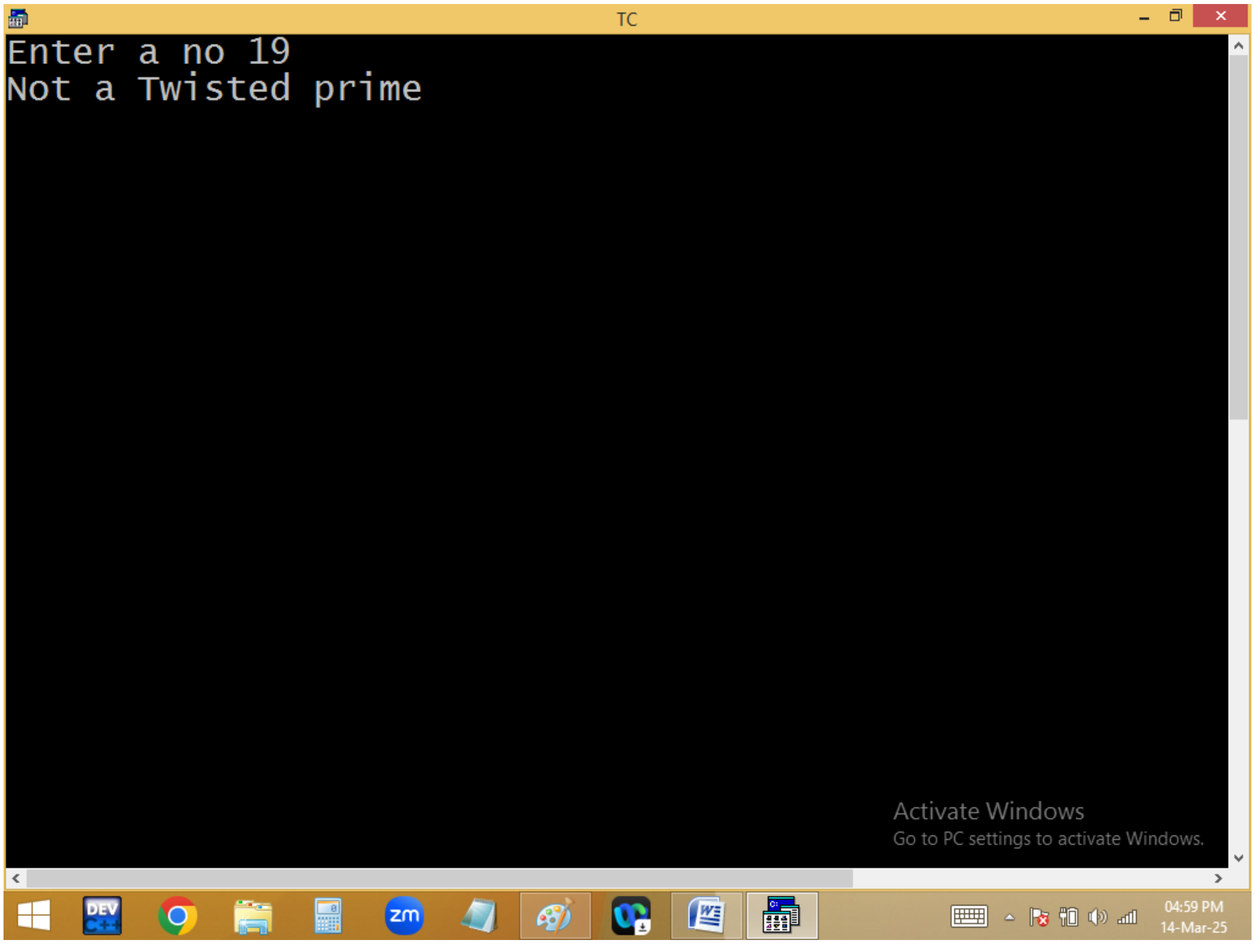
Finding twisted prime or not

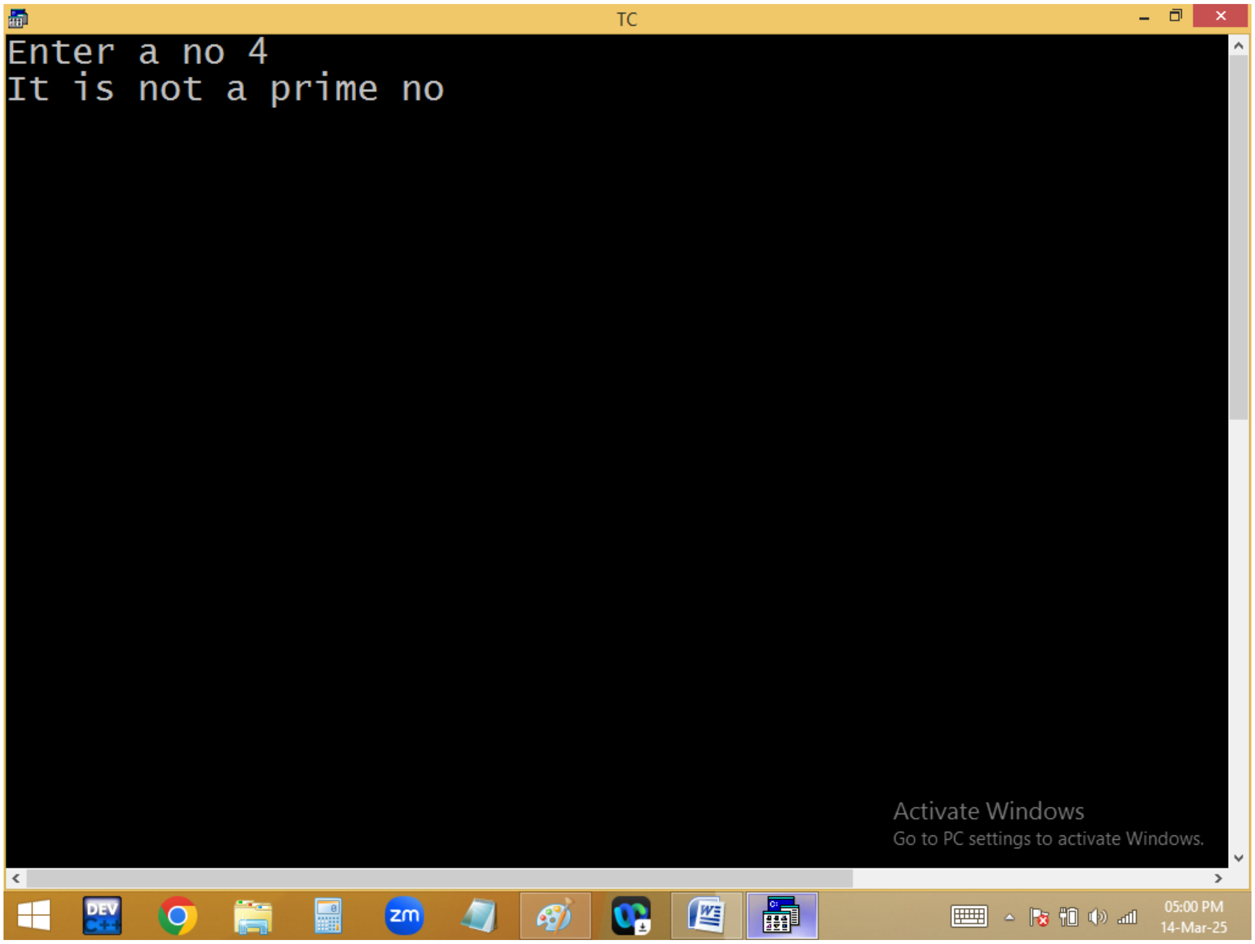
13 is a prime and 13 reverse 31 also prime


TC
File Edit Run Compile Project Options Deb
Line 5 Col 14 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,rev=0,c=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(i=1;i<=n;i++)if(n%i==0)c++;
if(c==2)
{
while(n){rev=rev*10+(n%10);n=n/10;}
for(c=0, i=1;i<=rev;i++)if(rev%i==0)c++;
puts(c==2?"It is a twisted prime":"Not a Twisted prime")
}
else puts("It is not a prime no");
getch();
}
Activate Windows
Go to PC settings to activate Windows.
04:58 PM
14-Mar-25

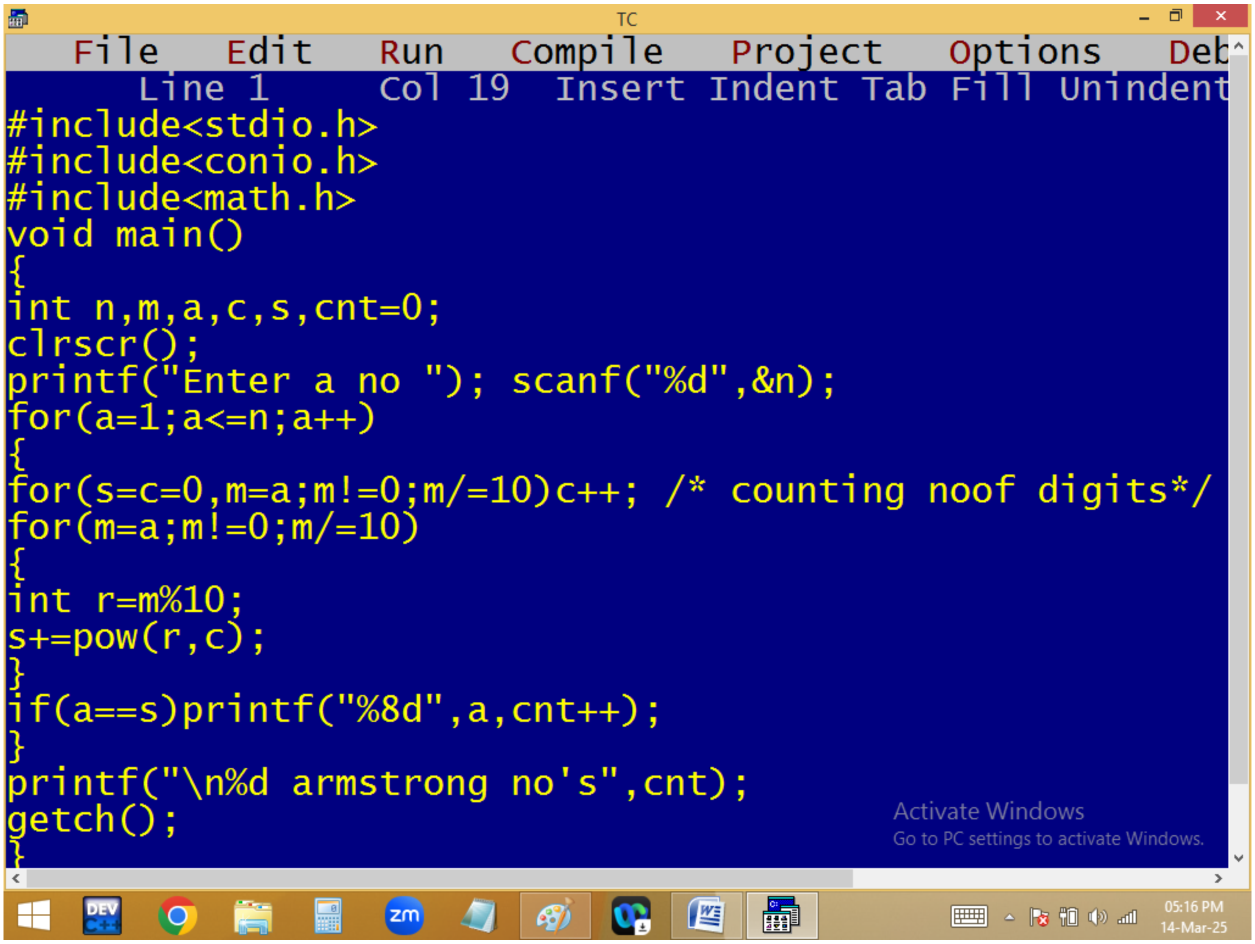

TC
Enter a no 31
It is a twisted prime
Activate Windows
Go to PC settings to activate Windows.
04:58 PM
14-Mar-25


TC
Enter a no 17
It is a twisted prime
Activate Windows
Go to PC settings to activate Windows.
04:58 PM
14-Mar-25

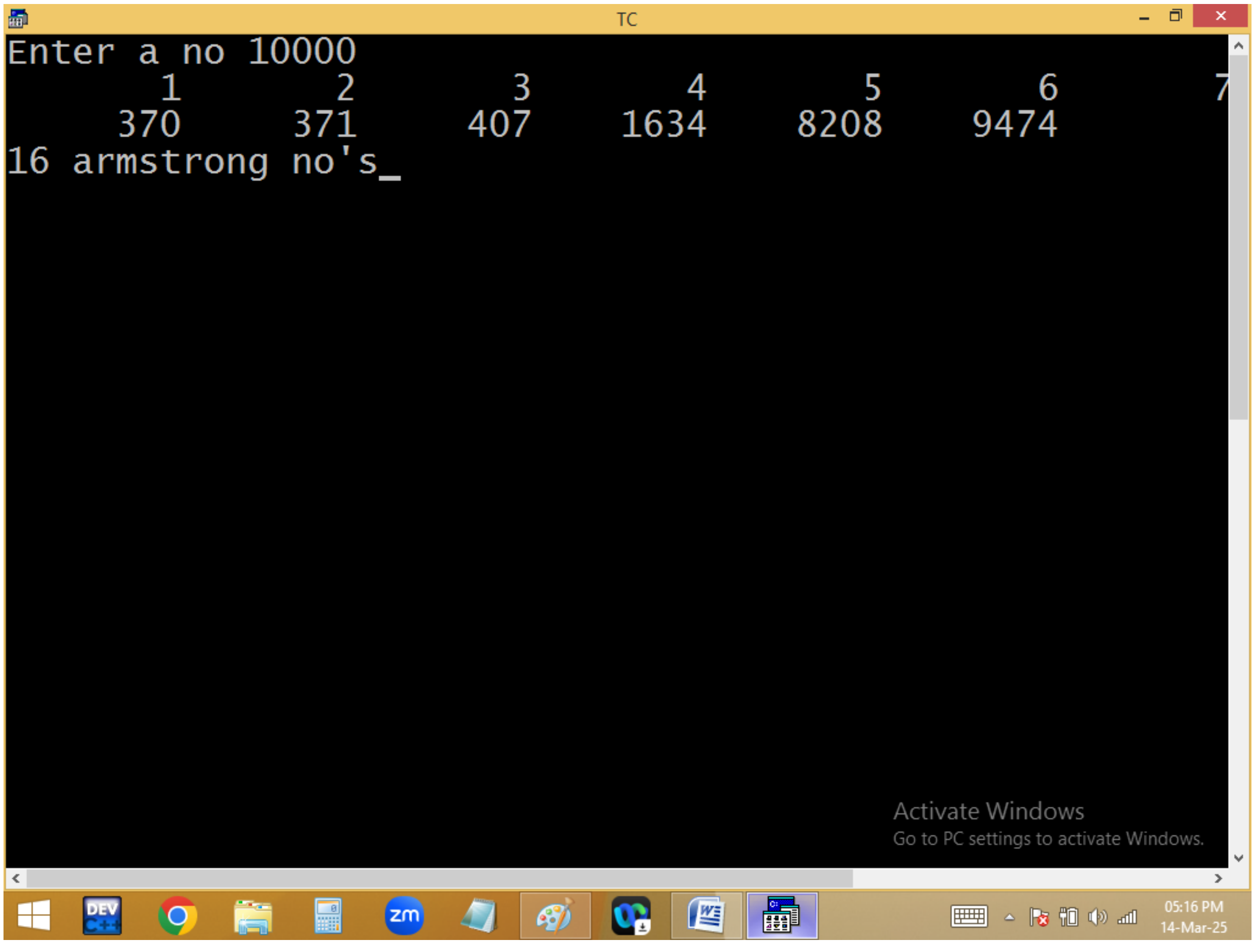

TC
Enter a no 19
Not a Twisted prime
Activate Windows
Go to PC settings to activate Windows.
04:59 PM
14-Mar-25



Printing 1..n Armstrong no’s:




```c
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int n,m,a,c,s,cnt=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(a=1;a<=n;a++)
{
for(s=c=0,m=a;m!=0;m/=10)c++; /* counting noof digits*/
for(m=a;m!=0;m/=10)
{
int r=m%10;
s+=pow(r,c);
}
if(a==s)printf("%8d",a,cnt++);
}
printf("\n%d armstrong no's",cnt);
getch();
}
```




```
              500
for( a=1; a<=n; a++ )
{     c=0;
for( m=a; m!=0; m/=10)c++;
s=0;
for( m=a; m!=0; m/=10 )
{
r=m%10;
s+=pow(r,c);
}
if(a==s)p(arm);
```

n

$\cancel{153} \% 10 = 3^3 = 27$

$\cancel{15} \% 10 = 5^3 = 125$

$1 \% 10 = 1^3 = 1$

153

112334

2

4

56525

6

2

11112